Week 7: MEC6602E Transonic Aerodynamics

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This week

- Convergence monitoring and running strategy;
- Post-processing;
- Verification and validation;
- Time-accurate simulation.

Convergence monitoring

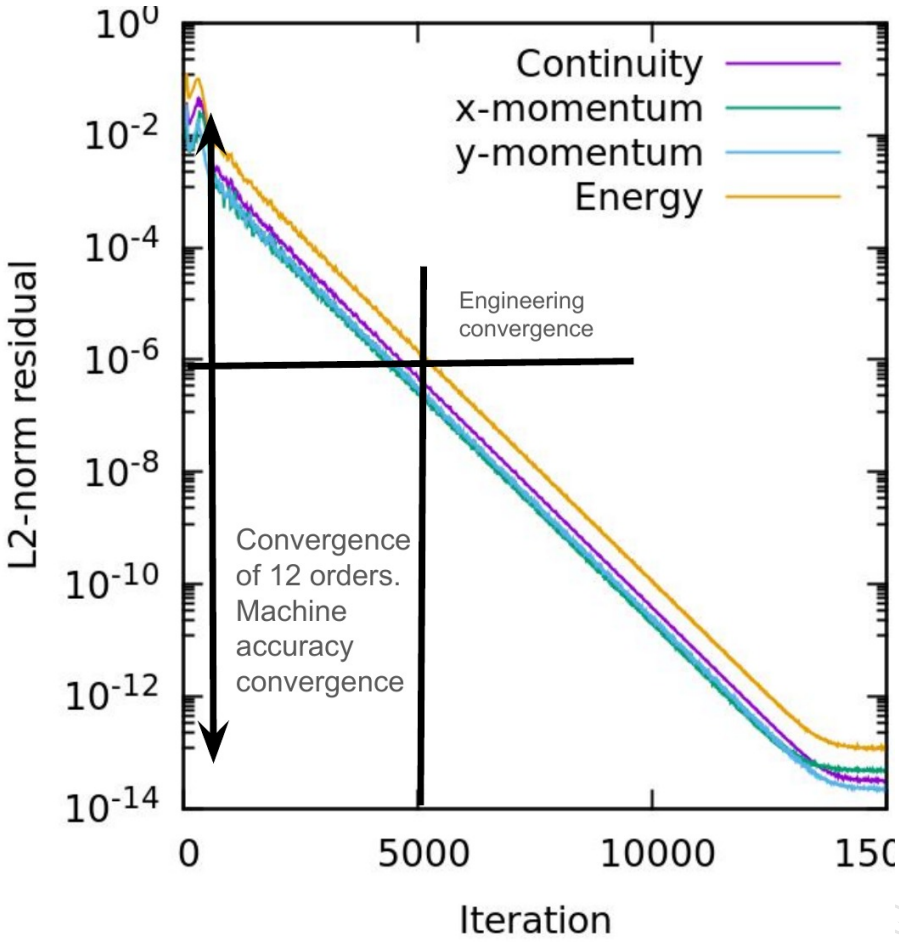
Residual convergence monitoring

Since the change of the solution per iteration is proportional to the sum of the fluxes (residual), we monitor the convergence with the L2-norm of the density residual vector.

Add to the code an exit criterion based on this quantity.

Print this value in a text file and plot it on a semilog scale.

Do not do the plot in the same script as the computation. This will destroy the performance.



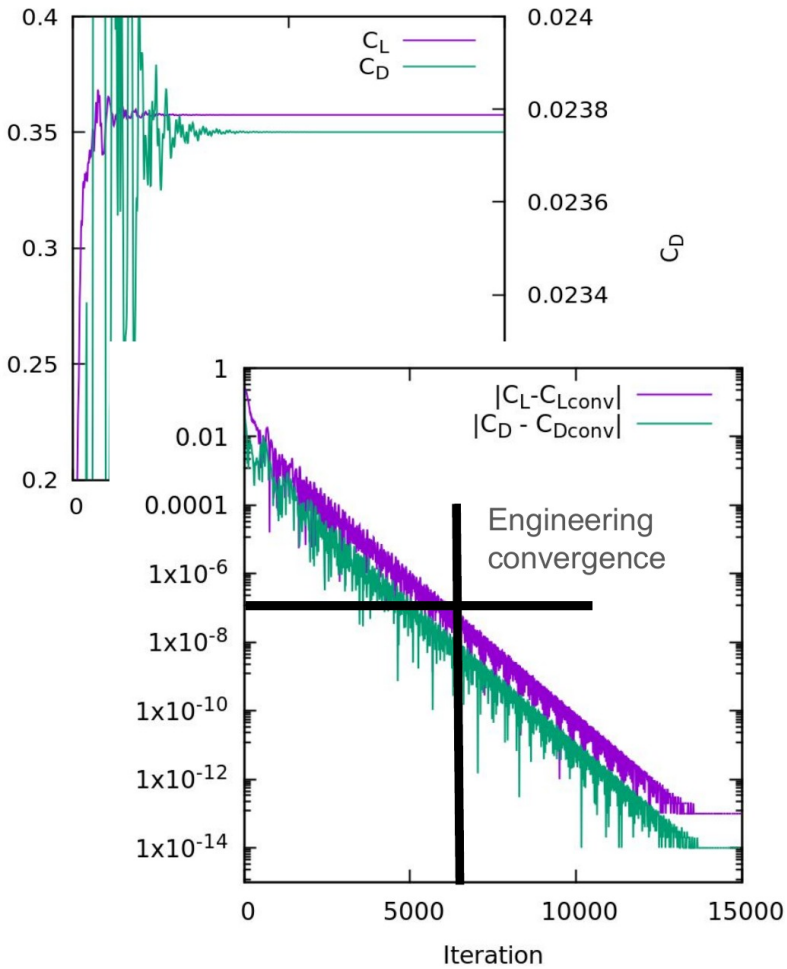
Force convergence monitoring

We are usually interested in the force and moment coefficients. Hence, we can monitor them and put an exit criterion based on the force and moment.

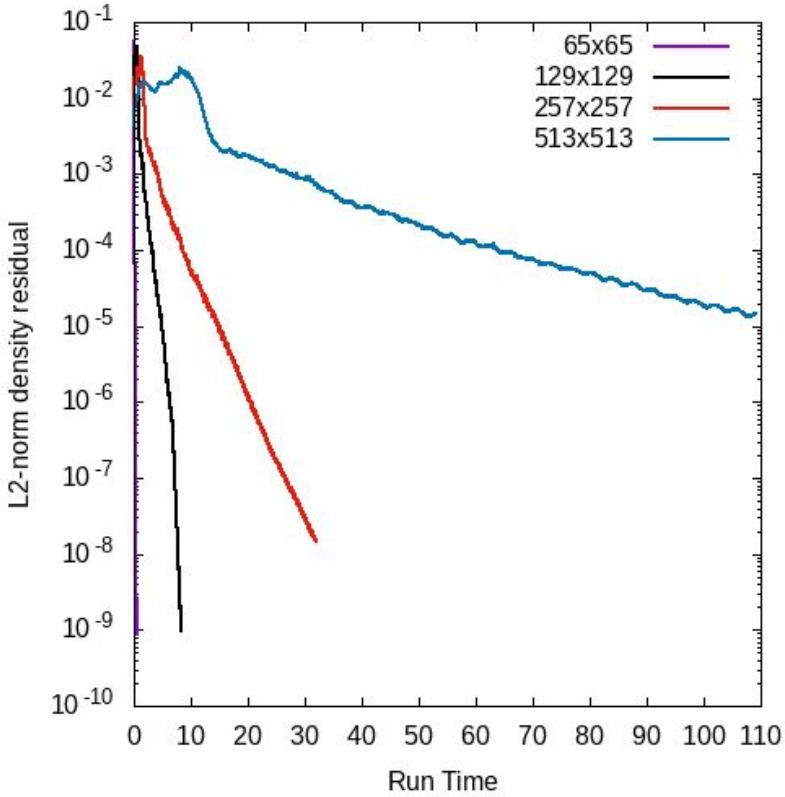
Typically we use the Cauchy criterion: if for all iterations m, n after N iterations

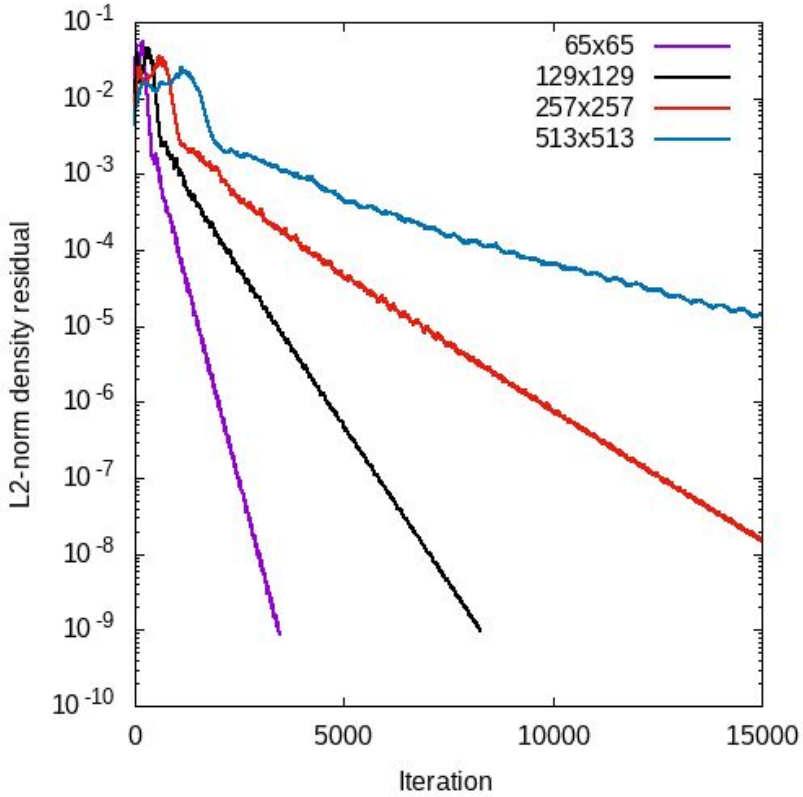
$$|\mathcal{F}_m - \mathcal{F}_n| < \varepsilon,$$

with ε a small number, exit the simulation if this is true for a given number of iterations.



Computation convergence on several grids





Convergence monitoring

It is good to verify that machine-accuracy convergence can be achieved, to verify that there are no errors in the implementation.

In applications, we do not reach machine-accuracy convergence:

- Because it costs computational power without added value;
- Because convergence stalls on complex cases.

We often talk of "engineering convergence" to describe a level of convergence suitable for the application.

Converging the force and moment coefficients is enough for external aerodynamic applications.

Running strategy

Initialization:

- "Cold start": start from the freestream conditions;
- "Warm start": start from a previously converged flow condition (typically another angle of attack).

In aircract application we often want to cover a full range of angle of attack (a polar). It is more efficient to run with warm start.

In your code, you can add a function to read a flow output file (like the Tecplot ASCII format). This enables starting from a converged field and can be useful if a run is not properly converged.

Lift coefficient targeting

Will be useful for the last homework

It is often more practical to have results at a specific C_L (an aircraft fly with a given weight, at a given altitude and speed as a given C_L) instead of a specific angle of attack. There are two possible ways:

- Compute at iso-AoA and then interpolate at the specific C_L:
  - Beware of the granularity of the parameter;
  - Be careful with the interpolation (C_D is non-linear with respect to C_L).
- Compute at iso-C_L:
  - More accurate for the targeted point;
  - Requires the implementation of an iterative method.

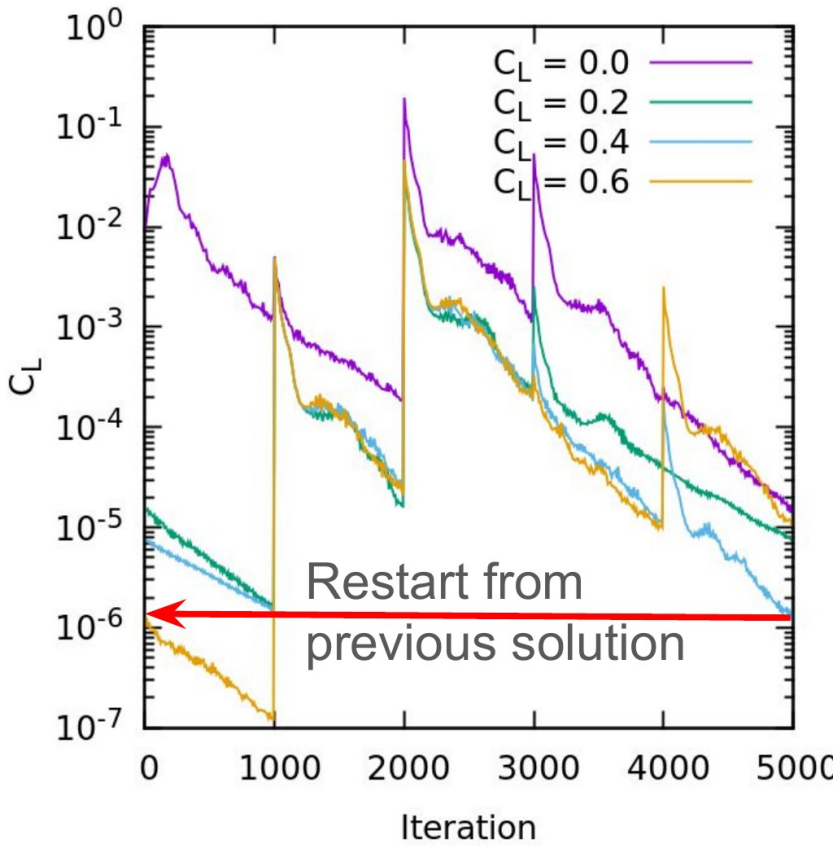
Iso-C_L computation

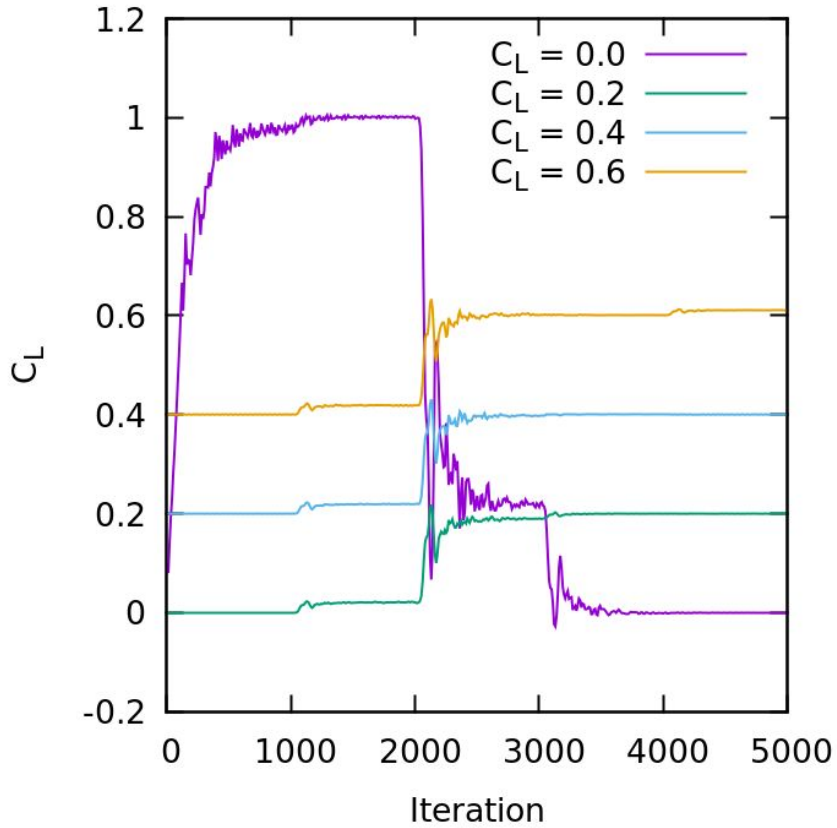
1. Initialize at a given AoA α;
2. Enter the iteration loop;
3. Compute to "convergence";
4. Store α_{old} and $C_{L,old}$;
5. Perturb the AoA by a small amount;
6. Compute to "convergence";
7. Iterate (when "converged"):
   a. $\Delta\alpha = \dfrac{(C_L - C_{L,t})(\alpha - \alpha_{old})}{C_L - C_{L,old}}$;
   b. Store α_{old} and $C_{L,old}$;
   c. $\alpha = \alpha + \Delta\alpha$.
8. Iterate to converge the flow (Euler solver) and the AoA (inner iteration); the AoA change is fed through the far-field BC.

Do not fully converge before changing the angle of attack.

Convergence with C_L targeting

Will be useful for the last homework





Post-processing

Post-processing

When post-processing the simulations, we are interested in:

- Force and moment coefficients;
- Pressure and friction (in viscous flow) coefficient distributions;
- Spanloads (distribution of the force and moment coefficients for a 3D case);
- Drag decomposition:
  - Pressure and friction drag;
  - Induced, wave, viscous pressure and friction drag.

Equation non-dimensionalization

We can show with the Buckingham Pi theorem that if a physical relation has N physical quantities which involve K fundamental dimensions (mass, length and time in classical fluid mechanics), there are $N - K$ non-dimensional quantities which define the problem.

In our case, let R be a functional like the lift, drag or moment:

$$R = f(\rho_\infty, V_\infty, c, \mu_\infty, a_\infty)$$

Then we find $6 - 3 = 3$ non-dimensional quantities:

$$C_R = f_6(\mathrm{Re}, M_\infty)$$

Ref: Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill.

Equation non-dimensionalization

- $C_R = R/\left(\frac{1}{2}\rho_\infty V_\infty^2 S_{ref}\right)$: the force coefficients (moments need a different non-dimensionalization);
- $\mathrm{Re} = \rho_\infty V_\infty c/\mu_\infty$: the Reynolds number (not involved in inviscid flow);
- $M_\infty = V_\infty/a_\infty$: the Mach number.

We can add a dependence on angles (which are by themselves non-dimensional).

Following the Buckingham Pi theorem, we can find other non-dimensional coefficients for different functionals (pressure coefficient, friction coefficient, moment coefficient, etc.).

Ref: Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill.

Equation non-dimensionalization

- Thus, the non-dimensional outputs are the same for a given set of non-dimensional parameters;
- This allows us to compare results between CFD, wind tunnel and flight test;
- This is why CFD codes are often non-dimensional.

Note: for multi-physics simulations, reducing a problem to non-dimensional quantities may be complex, and it is sometimes more practical to solve the equations with physical dimensions.

Ref: Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill.

Force computation

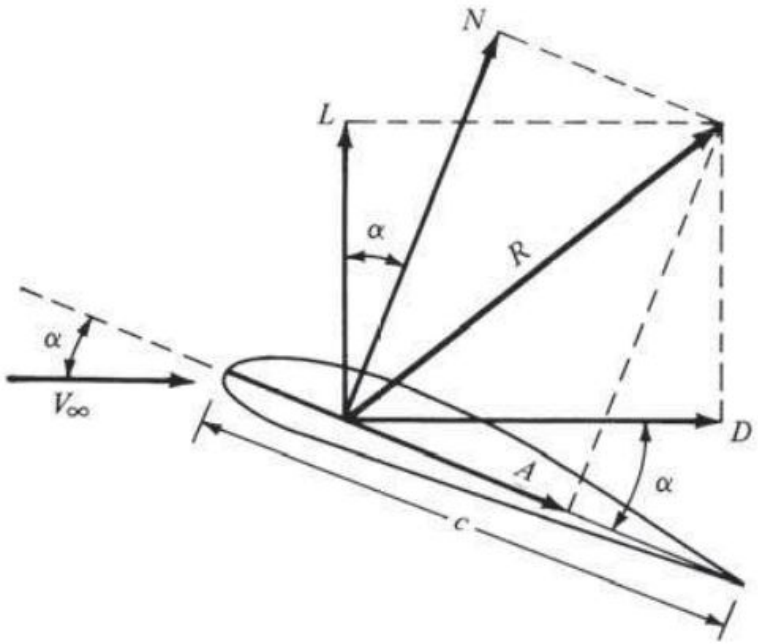


Figure 1.17 Resultant aerodynamic force and the components into which it splits.

Lift and drag coefficients:

$$c_l = c_n \cos\alpha - c_a \sin\alpha$$

$$c_d = c_n \sin\alpha + c_a \cos\alpha$$

Lift coefficient: $C_L \equiv \dfrac{L}{q_\infty S}$

Drag coefficient: $C_D \equiv \dfrac{D}{q_\infty S}$

Normal force coefficient: $C_N \equiv \dfrac{N}{q_\infty S}$

Axial force coefficient: $C_A \equiv \dfrac{A}{q_\infty S}$

Moment coefficient: $C_M \equiv \dfrac{M}{q_\infty S l}$ Pressu

Ref: Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill.

Force computation

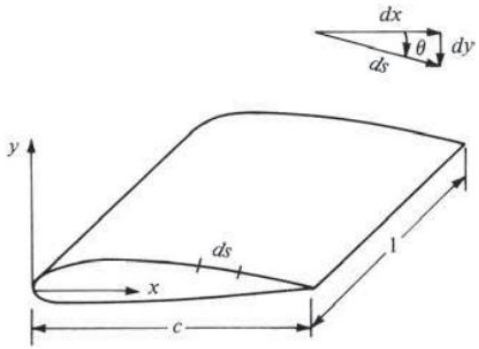


Figure 1.22 Geometrical relationship of differential lengths.

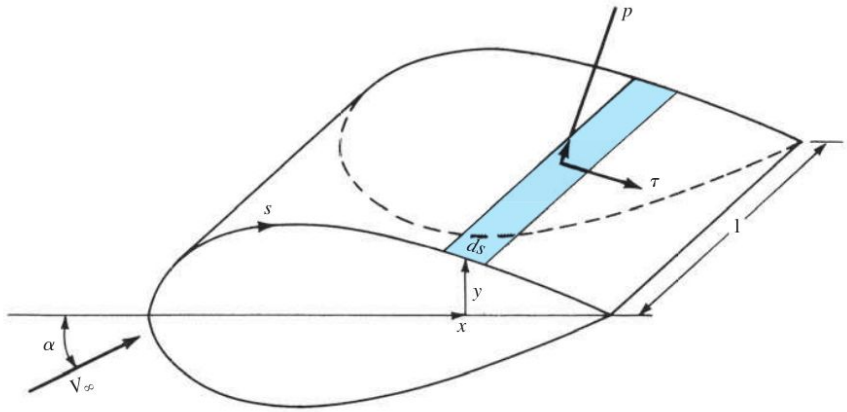


Figure 1.19 Aerodynamic force on an element of the body surface.

$$N' = -\int_{LE}^{TE} (p_u \cos\theta + \tau_u \sin\theta)\, ds_u + \int_{LE}^{TE} (p_l \cos\theta - \tau_l \sin\theta)\, ds_l$$

$$A' = \int_{LE}^{TE} (-p_u \sin\theta + \tau_u \cos\theta)\, ds_u + \int_{LE}^{TE} (p_l \sin\theta + \tau_l \cos\theta)\, ds_l$$

Ref: Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill.

Force computation

$$c_n = \frac{1}{c}\left[\int_0^c \left(C_{p,l} - C_{p,u}\right) \mathrm{d}x + \int_0^c \left(c_{f,u}\frac{\mathrm{d}y_u}{\mathrm{d}x} + c_{f,l}\frac{\mathrm{d}y_l}{\mathrm{d}x}\right) \mathrm{d}x\right]$$

$$c_a = \frac{1}{c}\left[\int_0^c \left(C_{p,u}\frac{\mathrm{d}y_u}{\mathrm{d}x} - C_{p,l}\frac{\mathrm{d}y_l}{\mathrm{d}x}\right) \mathrm{d}x + \int_0^c \left(c_{f,u} + c_{f,l}\right) \mathrm{d}x\right]$$

$$c_{m_{LE}} = \frac{1}{c^2}\left[\int_0^c \left(C_{p,u} - C_{p,l}\right) x \, \mathrm{d}x - \int_0^c \left(c_{f,u}\frac{\mathrm{d}y_u}{\mathrm{d}x} + c_{f,l}\frac{\mathrm{d}y_l}{\mathrm{d}x}\right) x \, \mathrm{d}x + \int_0^c \left(C_{p,u}\frac{\mathrm{d}y_u}{\mathrm{d}x} + c_{f,u}\right) y_u \, \mathrm{d}x + \int_0^c \left(-C_{p,l}\frac{\mathrm{d}y_l}{\mathrm{d}x} + c_{f,l}\right) y_l \, \mathrm{d}x\right]$$

We usually compute the moment coefficient around the quarter chord. This can simply be done by a force and moment balance.

Numerically, the integrals are replaced by summations over the wall faces.

Ref: Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill.

Drag decomposition

The simplest drag decomposition (near-field) is:

- Pressure drag (integrate the pressure over the surface);
- Friction drag (integrate the friction over the surface).

For a more detailed analysis, we want a phenomenological decomposition (far-field):

- Induced drag (a part of the pressure drag, the cost of generating lift in 3D);
- Wave drag (caused by shock waves, also pressure drag);
- Viscous pressure drag (caused by boundary layers);
- Friction drag (same as in near-field).

Far-field drag decomposition

Not to be done in homework

The variation of entropy in the domain allows us to separate drag sources between:

- Irreversible sources: generate entropy (shock wave, boundary layer, wake);
- Reversible sources: no entropy generation (induced drag).

Volumes are identified around regions of drag generation, and the momentum conservation equation is integrated over these volumes.

This method allows us to identify the "spurious drag" from the numerical viscosity as the difference between the near-field and the far-field drags.

Spurious drag depends on the accuracy of the numerical scheme and on the grid.

Far-field drag decomposition

Not to be done in homework

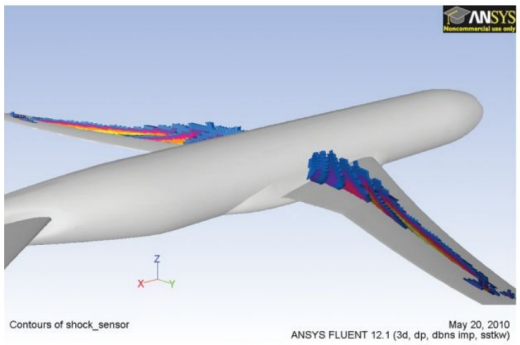


Fig. 3 Shock volume for the DLR F6 at $Ma = 0.85$ and $C_L = 0.5$.

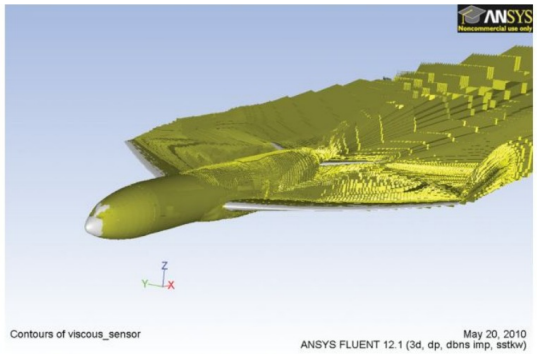


Fig. 4 Viscous volume for the DLR F6 at $Ma = 0.85$ and $C_L = 0.5$.

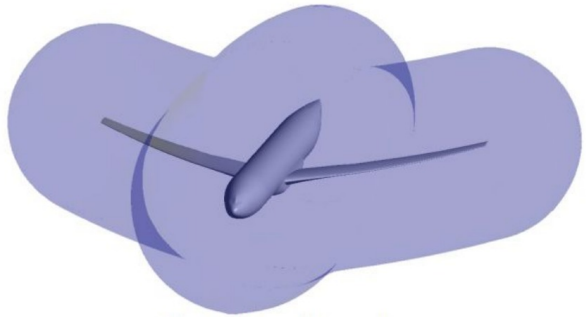
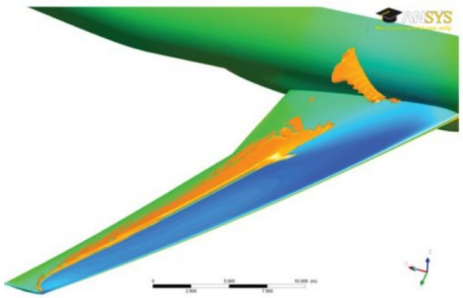


Fig. 5 Induced drag volume.



Ref: Gariépy, M., Malouin, B., Trépanier, J. Y., & Laurendeau, É. (2013). Far-field drag decomposition applied to the drag prediction workshop 5 cases. *Journal of Aircraft*, 50(6), 1822-1831.

Far-field correction

Not to be done in homework

In CFD simulations with a far-field boundary condition, the solution depends on the far-field distance. We can reduce this dependency by assuming that the airfoil behaves like a vortex and correcting the far-field condition (vortex correction).

$$u_\infty^\star = u_\infty + \left(\frac{\Gamma\sqrt{1-M_\infty^2}}{2\pi d}\right)\frac{\sin\theta}{1-M_\infty^2\sin^2(\theta-\alpha)}$$

$$v_\infty^\star = v_\infty - \left(\frac{\Gamma\sqrt{1-M_\infty^2}}{2\pi d}\right)\frac{\cos\theta}{1-M_\infty^2\sin^2(\theta-\alpha)}$$

$$p_\infty^\star = \left[p_\infty^{(\gamma-1)/\gamma} + \frac{\gamma-1}{2\gamma}\frac{\rho_\infty\left(\|\vec{v}_\infty\|_2^2 - \|\vec{v}_\infty^\star\|_2^2\right)}{p_\infty^{1/\gamma}}\right]^{\frac{\gamma}{\gamma-1}}$$

$$\rho_\infty^\star = \rho_\infty\left(\frac{p_\infty^\star}{p_\infty}\right)^{1/\gamma}$$

$$\Gamma = \tfrac{1}{2}\|\vec{v}_\infty\|_2\, c\, C_L$$

(Kutta-Joukowski theorem)

$$d = \sqrt{(x-x_{ref})^2 + (y-y_{ref})^2}$$

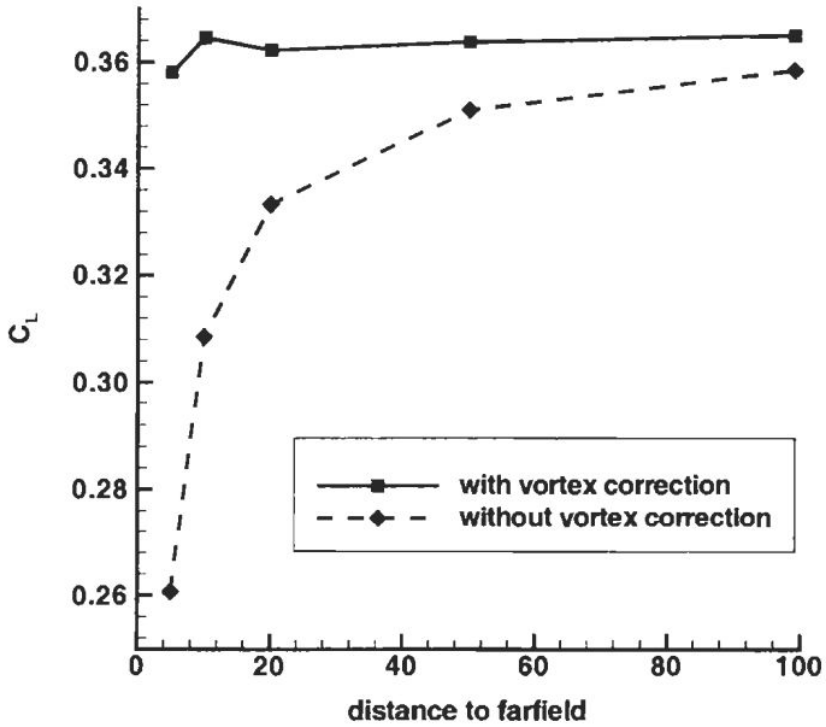
$$\theta = \arctan\left(\frac{y-y_{ref}}{x-x_{ref}}\right)$$

The correction is valid for subsonic flow and usable for transonic flow.

Ref: Blazek, J. (2015). *Computational fluid dynamics: principles and applications*. Butterworth-Heinemann.

Farfield correction

Not to be done in homework



The vortex correction allows us to limit the far-field distance to $50c$ or $100c$ and to reduce the number of grid points.

Without it, the far field must be very far ($500c$ or more).

In 3D, the correction is more complex and rarely used. However, full-aircraft simulations have a relief effect which makes them less dependent on the far-field size. A far field at a kilometre is usually sufficient.

Ref: Blazek, J. (2015). *Computational fluid dynamics: principles and applications*. Butterworth-Heinemann.

Verification and validation

Verification and validation

Verification: Does the software properly solve the equations?

- Does the method have the predicted order of accuracy?
- Are the solutions converging towards analytical solutions?
- Is the software consistent with other software solving the same equations?

Validation: Are the equations the correct ones to predict the physics?

- Comparison to experimental results;
- Is the mesh sufficiently refined to capture the physics?
- For the Euler equations: is the inviscid assumption acceptable?
- For the Navier-Stokes equations: is the turbulence model accurate for a given problem?

Verification - estimation of order of accuracy

$$(\mathcal{F}_f - \mathcal{F}_m)h_c^p + (\mathcal{F}_c - \mathcal{F}_f)h_m^p + (\mathcal{F}_m - \mathcal{F}_c)h_f^p = 0$$

$$h_c = \omega h_m = \omega^2 h_f = 2h_m = 4h_f \quad \text{if } \omega = 2$$

$$(\mathcal{F}_c - \mathcal{F}_f) = (\mathcal{F}_c - \mathcal{F}_m) + (\mathcal{F}_m - \mathcal{F}_f)$$

Combining:

$$\mathcal{R} = \frac{\mathcal{F}_f - \mathcal{F}_m}{\mathcal{F}_m - \mathcal{F}_c} = \frac{\omega^p - 1}{\omega^{2p} - \omega^p} = \frac{2^p - 1}{4^p - 2^p}$$

$\mathcal{F}$ is a functional like the lift, drag or moment coefficient (f: fine, m: medium, c: coarse mesh).

h is the grid spacing and ω is the ratio of cell sizes.

$\mathcal{R}$ is the ratio of the functional changes between two meshes; p is the order of accuracy.

We want to solve for the order of accuracy.

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

Verification - estimation of order of accuracy

$\mathcal{R}$ is known from a set of simulations on different grids. Thus, we can define a function of p for which we seek the zero:

$$\mathcal{G}(p) = (\omega^{2p} - \omega^p)\mathcal{R} - \omega^p + 1 = (4^p - 2^p)\mathcal{R} - 2^p + 1 = 0$$

We can use a Newton method to find the zero of this non-linear function. We need the derivative of $\mathcal{G}$ with respect to p:

$$\begin{aligned}\mathcal{G}_p(p) &= \ln(\omega^2)\,\omega^{2p}\mathcal{R} - \ln(\omega)\,\omega^p(\mathcal{R}+1) \\ &= \ln(4)\,4^p\mathcal{R} - \ln(2)\,2^p(\mathcal{R}+1) \qquad \text{if } \omega = 2\end{aligned}$$

Then we can solve for p iteratively as:

$$p^{n+1} = p^n - \frac{\mathcal{G}(p^n)}{\mathcal{G}_p(p^n)}$$

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.



Verification - estimation of order of accuracy

Once we know the order of accuracy, we can estimate the value as the mesh spacing goes to zero with:

$$\mathcal{F}^\star = \mathcal{F}_f + \frac{\mathcal{F}_f - \mathcal{F}_m}{\omega^p - 1} = \mathcal{F}_f + \frac{\mathcal{F}_f - \mathcal{F}_m}{2^p - 1}$$

In the specific case of the inviscid flow around an airfoil without a shock wave (subcritical flow), the drag should be zero. Thus, the drag coefficient is the error:

$$C_d^\star = 0, \qquad \bar{p} = \log_\omega \left[\frac{C_{d_m}}{C_{d_f}}\right] = \log_2 \left[\frac{C_{d_m}}{C_{d_f}}\right]$$

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

For your homework, do analysis for at least one

FLO82 order of accuracy

Table 2 FLO82-H-CUSP subcritical data at $M = 0.5$

Mesh		$\alpha = 0$ deg		$\alpha = 1.25$ deg	
NC	NC^2	C_d	C_l	C_d	C_m
32	1,024	+0.001979418	+0.181639135	+0.002482085	−0.002926241
64	4,096	+0.002270216	+0.178822451	+0.001974608	−0.002487849
128	16,384	+0.000307646	+0.180014345	+0.000298397	−0.002294411
256	65,536	−0.000013299	+0.180458183	−0.000011045	−0.002301404
512	262,144	−0.000015191	+0.180446183	−0.000015534	−0.002290609
1,024	1,048,576	−0.000005076	+0.180382615	−0.000005301	−0.002276577
2,048	4,194,304	−0.000001404	+0.180354386	−0.000001481	−0.002270586
4,096	16,777,216	−0.000000365	+0.180347832	−0.000000388	−0.002269217
Continuum		+0.000000045	+0.180345850	+0.000000050	−0.002268812
Order p		1.821	2.107	1.805	2.130
$\bar{p}$		1.944	——	1.932	——

Table 3 FLO82-HCUSP transonic data at $M = 0.8$

Mesh		$\alpha = 0$ deg		$\alpha = 1.25$ deg	
NC	NC^2	C_d	C_l	C_d	C_m
32	1,024	+0.011451356	+0.387842746	+0.027861593	−0.042947499
64	4,096	+0.010264792	+0.372921380	+0.025487792	−0.043584427
128	16,384	+0.008500758	+0.373469550	+0.023786371	−0.043873739
256	65,536	+0.008312402	+0.368980205	+0.023357651	−0.042552941
512	262,144	+0.008328328	+0.363747900	+0.023084749	−0.041002228
1,024	1,048,576	+0.008338967	+0.360812844	+0.022934404	−0.040136414
2,048	4,194,304	+0.008341760	+0.358281928	+0.022799839	−0.039388829
4,096	16,777,216	+0.008342211	+0.357142338	+0.022737860	−0.039051466
Continuum		+0.008342298	+0.356208937	+0.022684938	−0.038774022
Order p		2.631	1.151	1.118	1.148

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

FLO82 order of accuracy

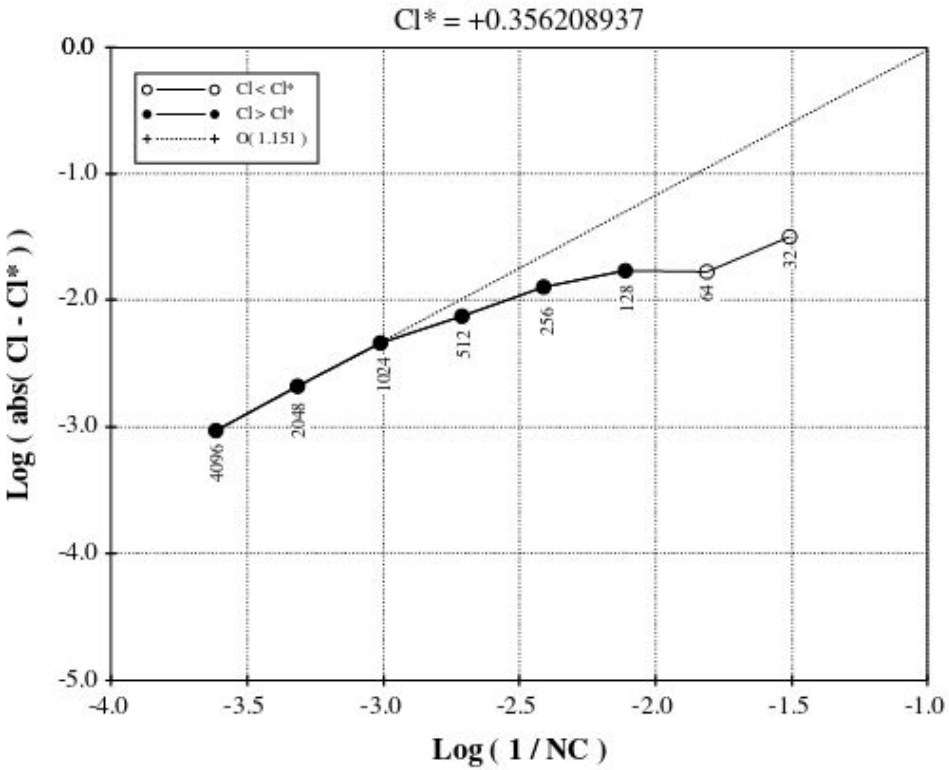


Fig. 23 FLO82 lift convergence $M = 0.80$, $\alpha = 1.25$ deg.

Lift convergence of FLO82 at $M = 0.80$, $\alpha = 1.25$ deg: error $|C_l - C_l^\star|$ against the cell size $1/NC$, in log-log scale.

The slope of the curve is the order of accuracy p.

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

NACA0012, AoA $= 0$ deg, $M = 0.5$

Reverse the C_p axis so that the suction side is on top of the graph

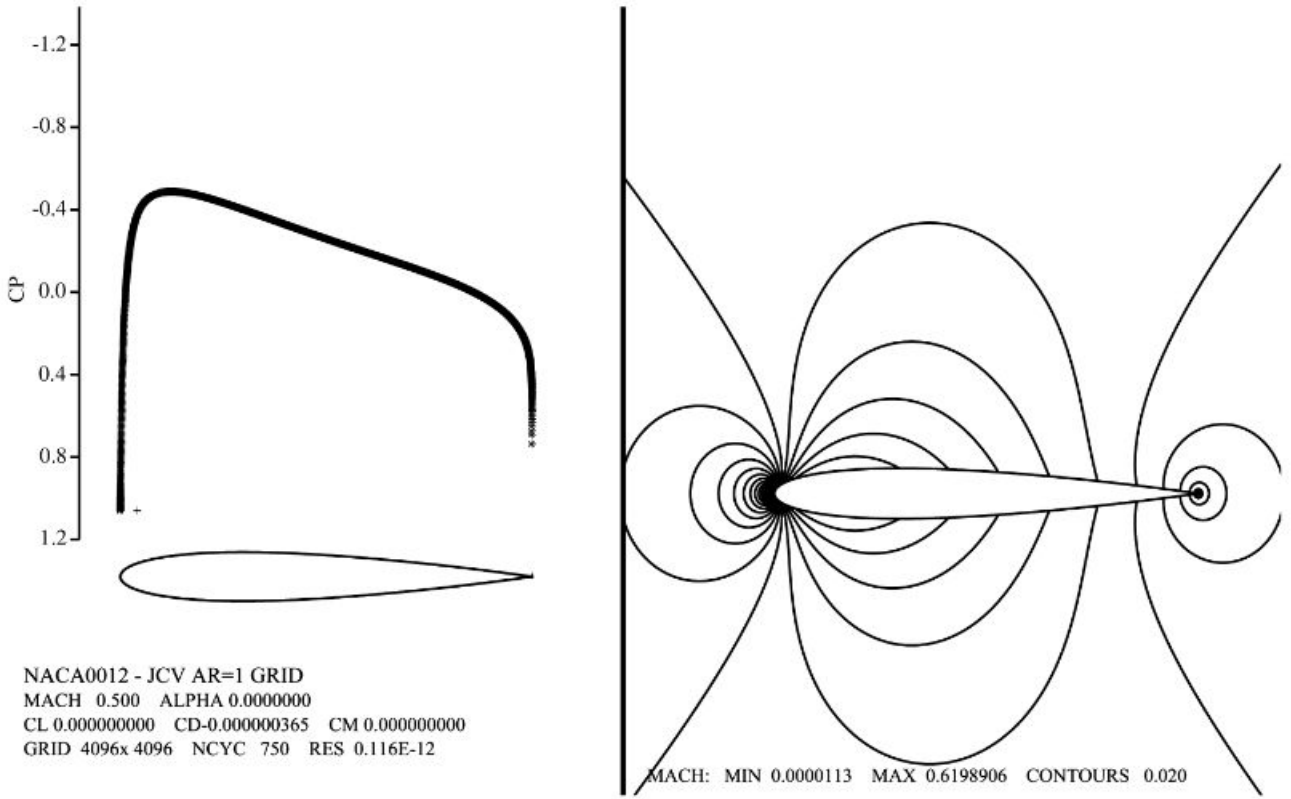


Fig. 13 FLO82 solution $M = 0.50, \alpha = 0.0$ deg.

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

NACA0012, AoA $= 0$ deg, $M = 0.8$

Reverse the C_p axis so that the suction side is on top of the graph

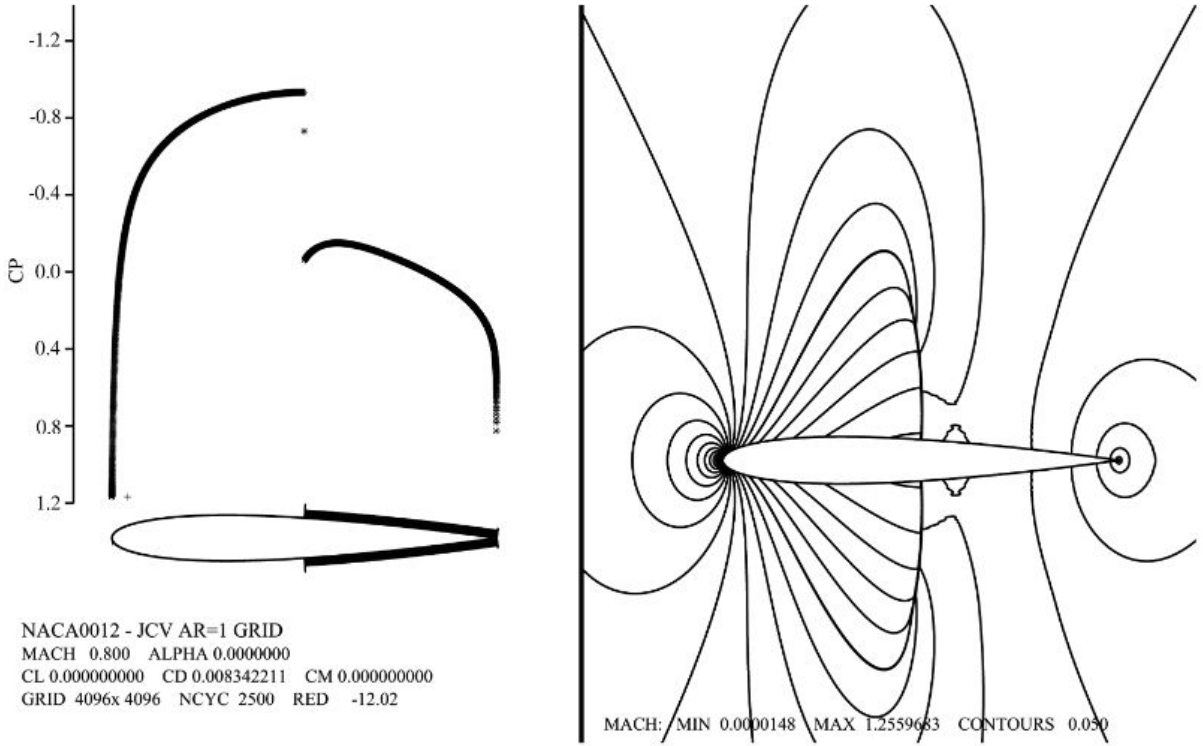


Fig. 15 FLO82 solution $M = 0.80, \alpha = 0.0$ deg.

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

NACA0012, AoA $= 1.25\,\text{deg}$, $M = 0.5$

Reverse the C_p axis so that the suction side is on top of the graph

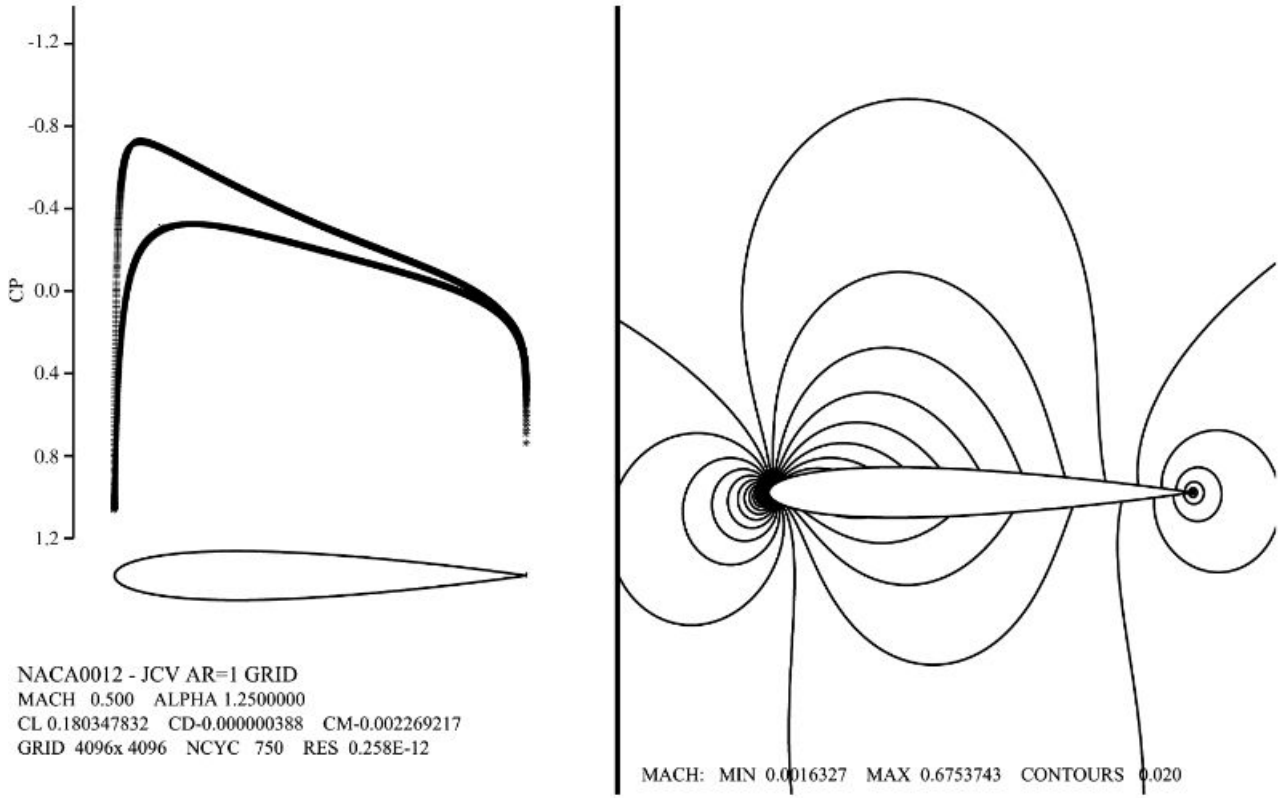


Fig. 14 FLO82 solution $M = 0.50, \alpha = 1.25\,\text{deg}$.

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

NACA0012, AoA $= 1.25\,\text{deg}$, $M = 0.8$

Reverse the C_p axis so that the suction side is on top of the graph

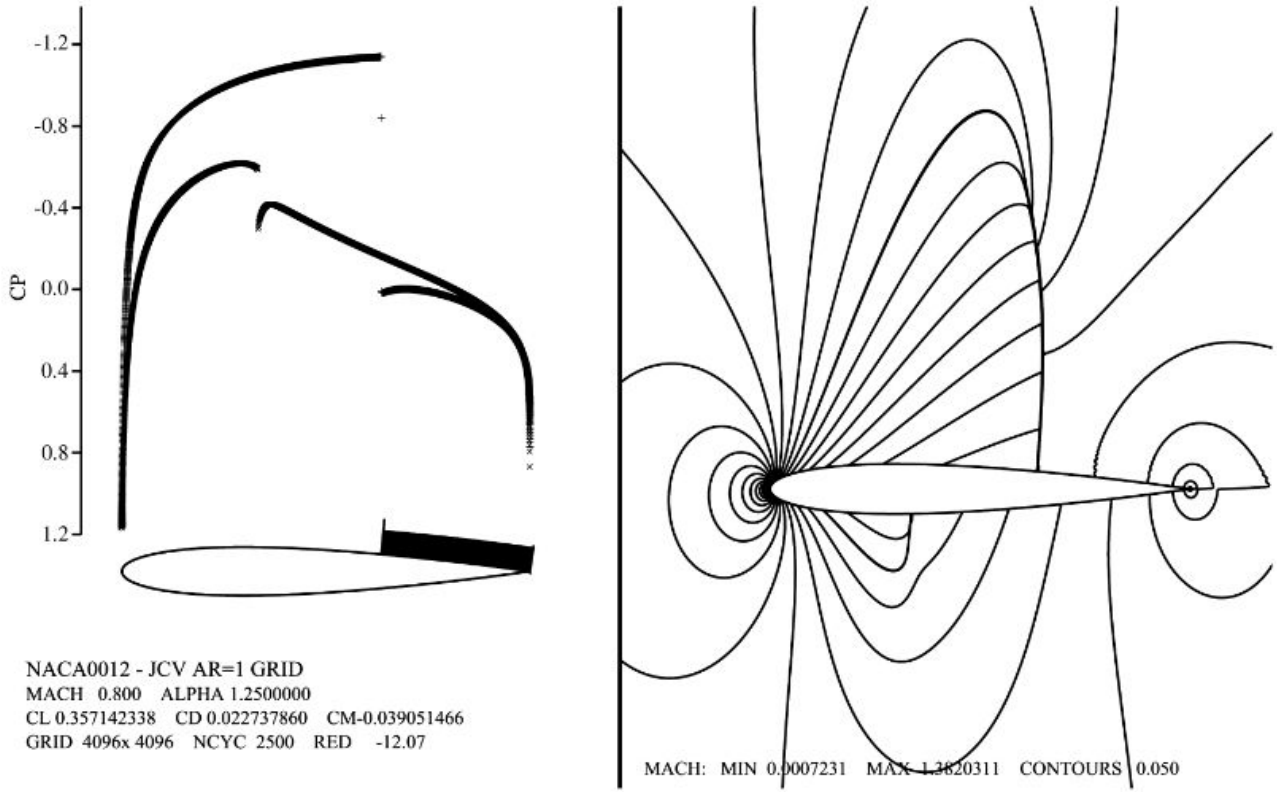
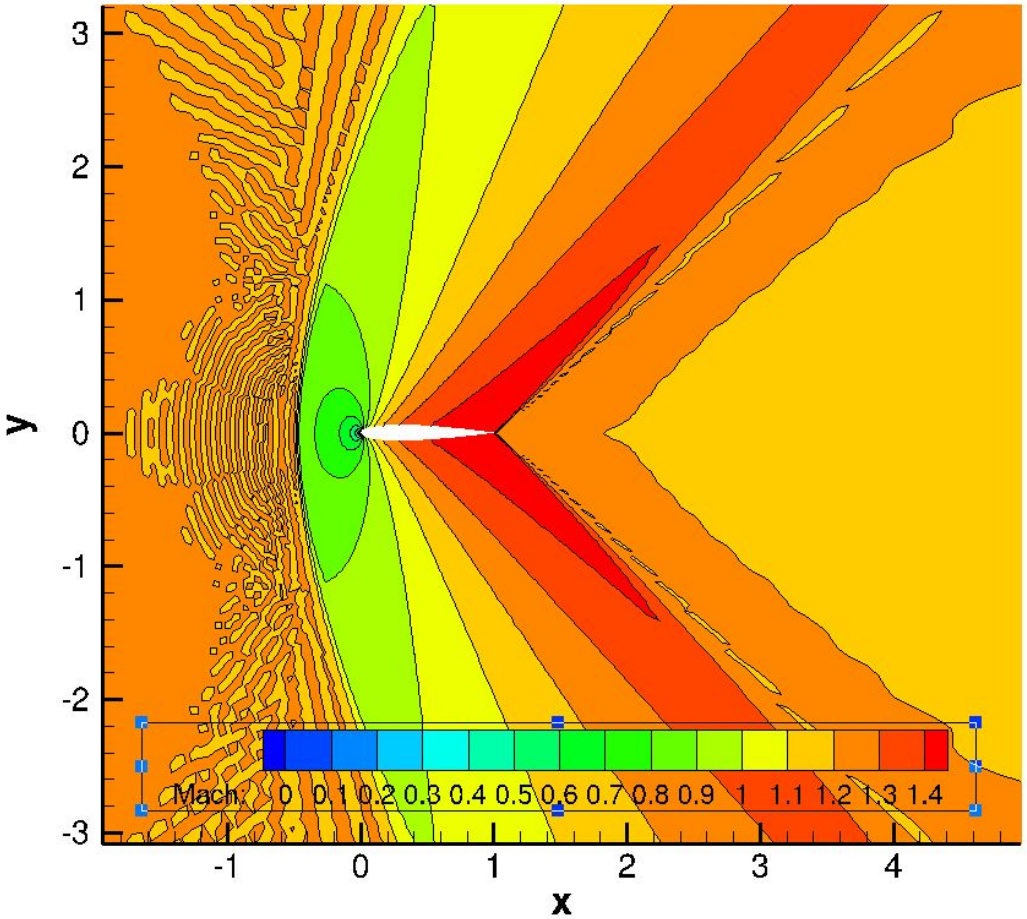


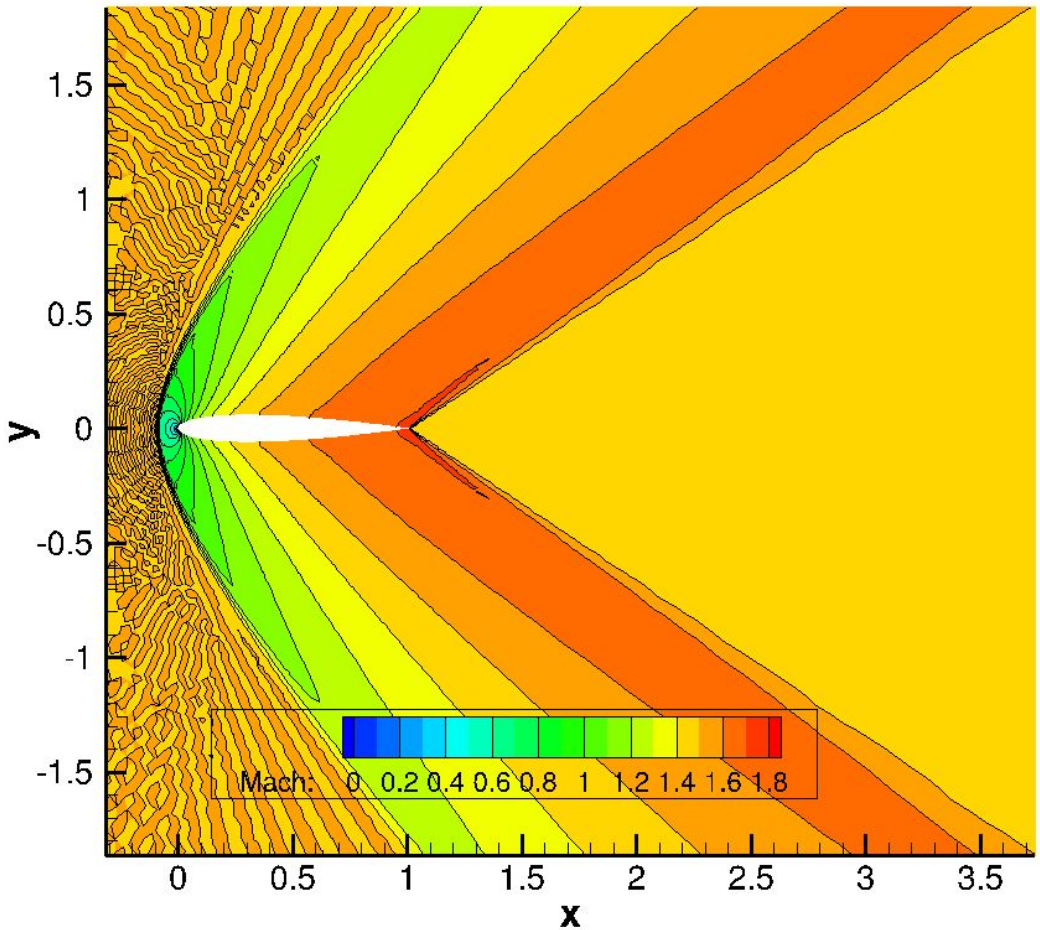
Fig. 16 FLO82 solution $M = 0.80, \alpha = 1.25\,\text{deg}$.

Ref: Vassberg, J. C., & Jameson, A. (2010). In pursuit of grid convergence for two-dimensional Euler solutions. *Journal of Aircraft*, 47(4), 1152-1166.

NACA0012, AoA $= 0$ deg, $M = 1.2$



NACA0012, AoA $= 0$ deg, $M = 1.5$

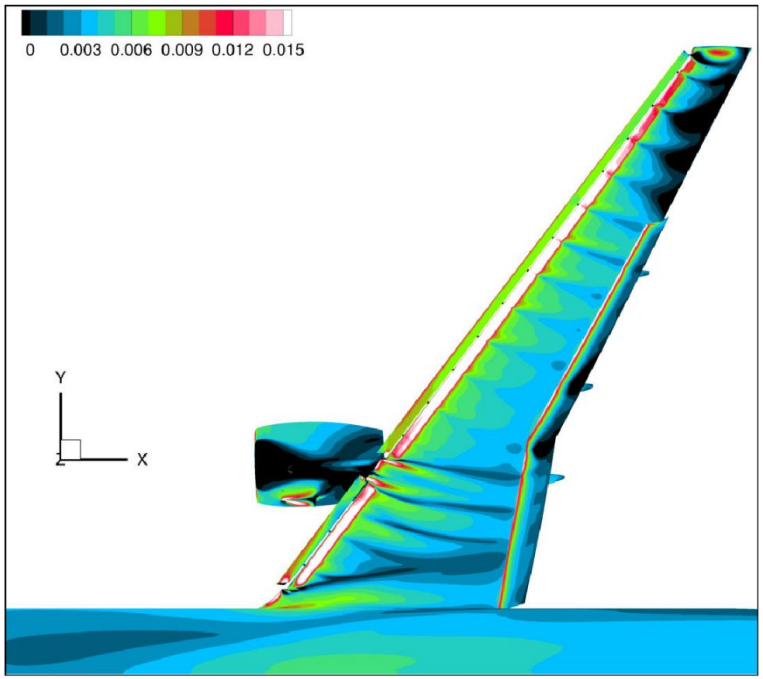


Verification and validation: HLPW4

The High-Lift Prediction Workshops are an international benchmark of state-of-the-art CFD codes for predicting the aircraft $C_{L_{max}}$.

Participants include:

- Laboratories (NASA, DLR, ONERA, etc.);
- Industry (Boeing, Airbus, Bombardier, Embraer, etc.);
- Universities;
- Vendors.



A set of test cases is submitted to participants:

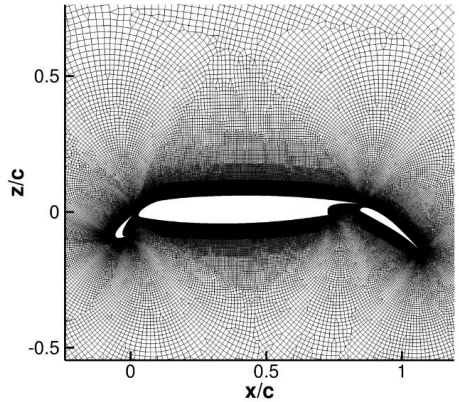
- 2D verification;
- Full-aircraft $C_{L_{max}}$;
- Sensitivity to flap deflection angles.

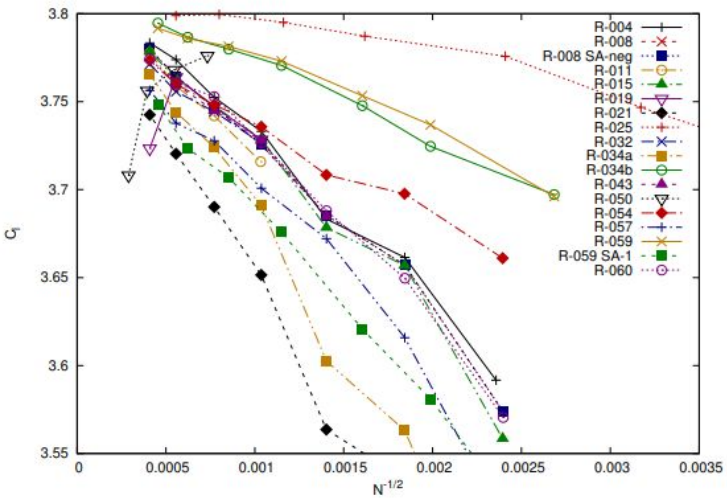
Ref: Ollivier Gooch, C. F., & Coder, J. G. (2022). Lessons learned by the fixed-grid RANS TFG for HLPW-4/GMGW-3. In *AIAA Aviation 2022 Forum* (p. 3211).

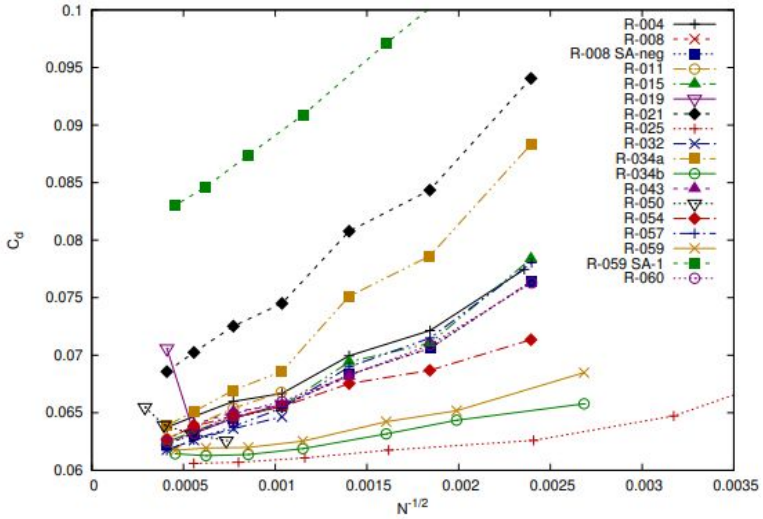
2D verification: HLPW4

Three-element airfoil at:

- AoA $= 16$ deg;
- $M = 0.2$;
- $\mathrm{Re} = 5$ million.



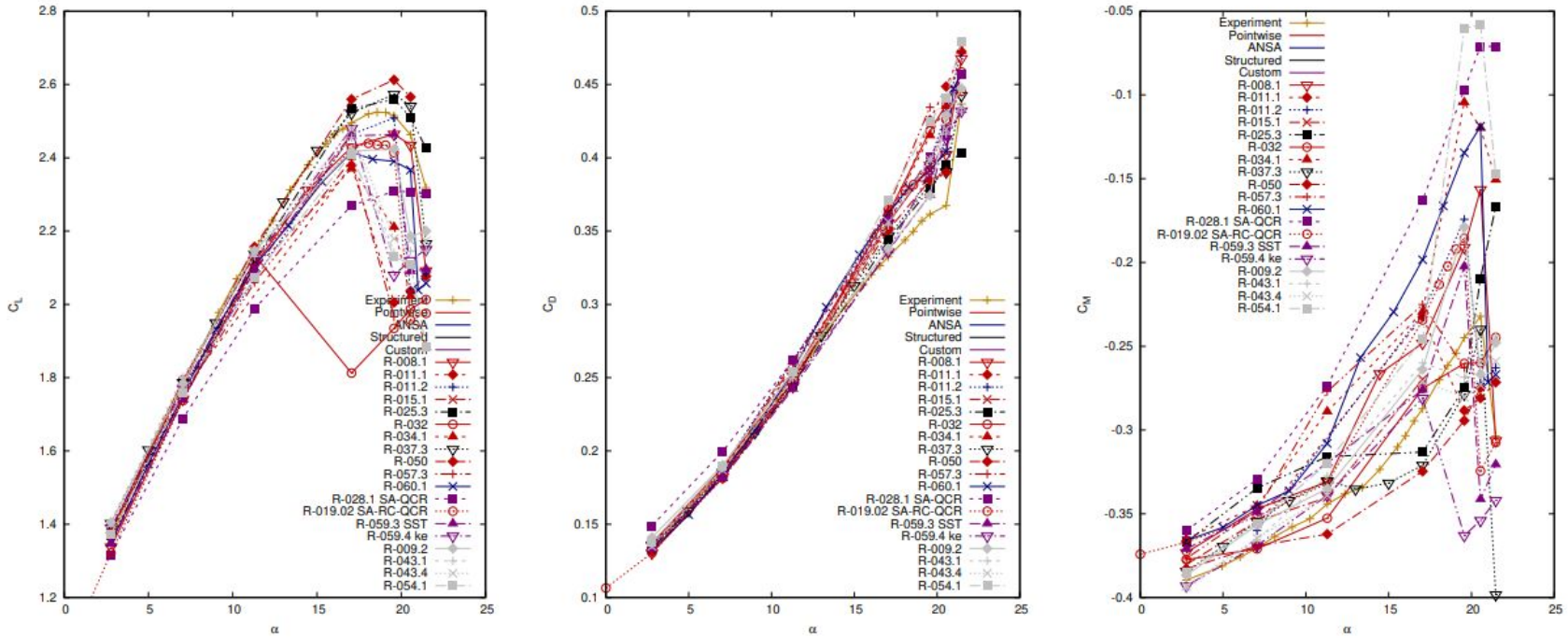




Ref: Ollivier Gooch, C. F., & Coder, J. G. (2022). Lessons learned by the fixed-grid RANS TFG for HLPW-4/GMGW-3. In *AIAA Aviation 2022 Forum* (p. 3211).

HLPW4 - $C_{L_{max}}$ prediction - all results

High-Lift CRM, $\mathrm{Re} = 5.49$ million, $M = 0.2$

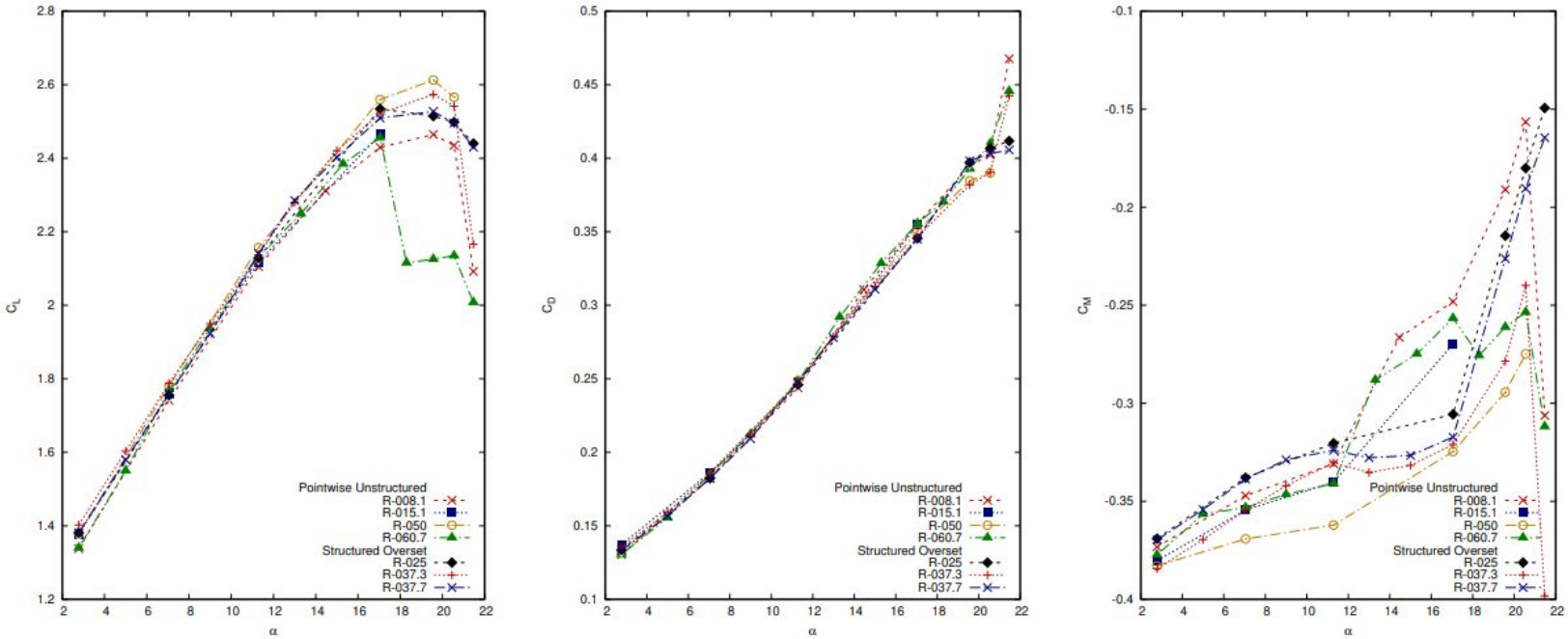


(a) Lift coefficient (b) Drag coefficient (c) Moment coefficient

Ref: Ollivier Gooch, C. F., & Coder, J. G. (2022). Lessons learned by the fixed-grid RANS TFG for HLPW-4/GMGW-3. In *AIAA Aviation 2022 Forum* (p. 3211).

HLPW4 - $C_{L_{max}}$ prediction - finest grids

High-Lift CRM, $\mathrm{Re} = 5.49$ million, $M = 0.2$

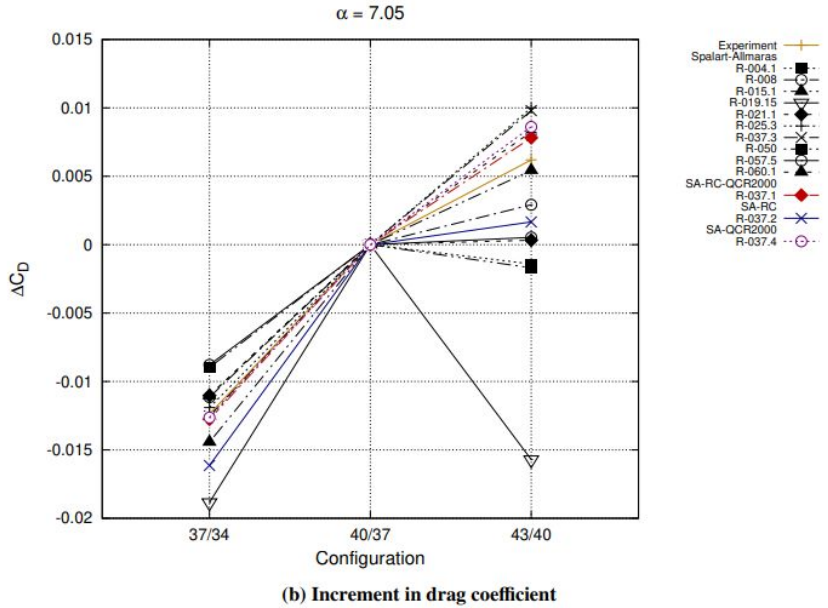


(a) Lift coefficient (b) Drag coefficient (c) Moment coefficient

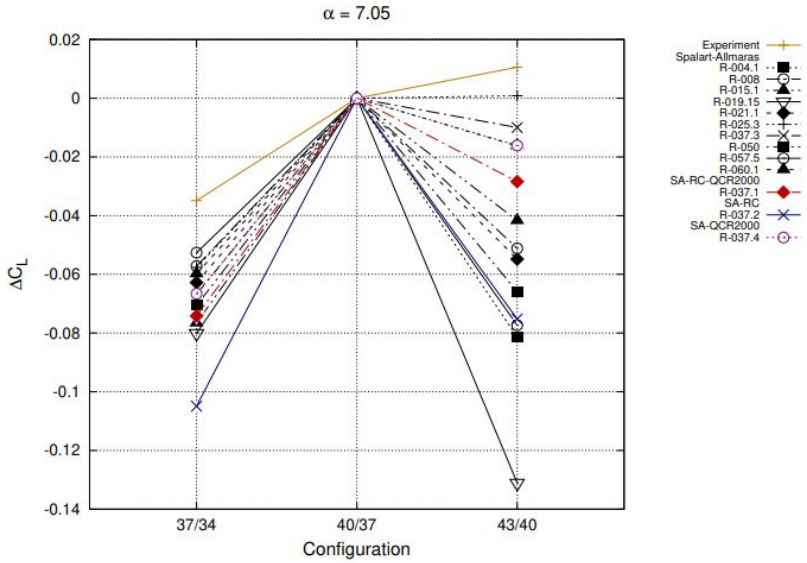
Ref: Ollivier Gooch, C. F., & Coder, J. G. (2022). Lessons learned by the fixed-grid RANS TFG for HLPW-4/GMGW-3. In *AIAA Aviation 2022 Forum* (p. 3211).

HLPW4 - flap effects

High-Lift CRM, three landing-flap deflections.

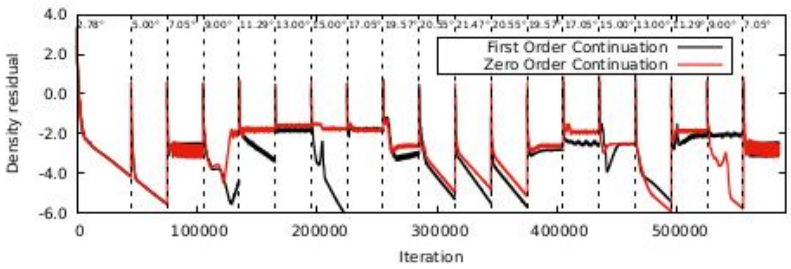


(b) Increment in drag coefficient

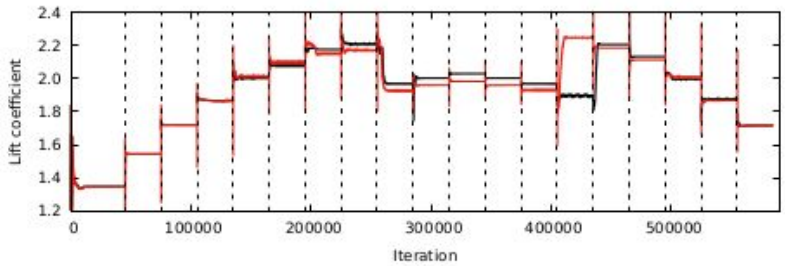


Ref: Ollivier Gooch, C. F., & Coder, J. G. (2022). Lessons learned by the fixed-grid RANS TFG for HLPW-4/GMGW-3. In *AIAA Aviation 2022 Forum* (p. 3211).

Warm restart



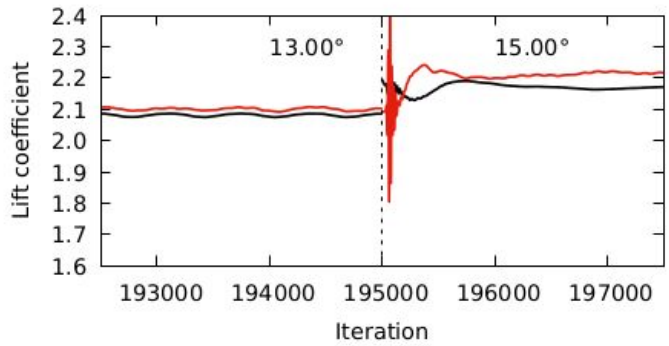
(a) Density residual



(b) Lift coefficient

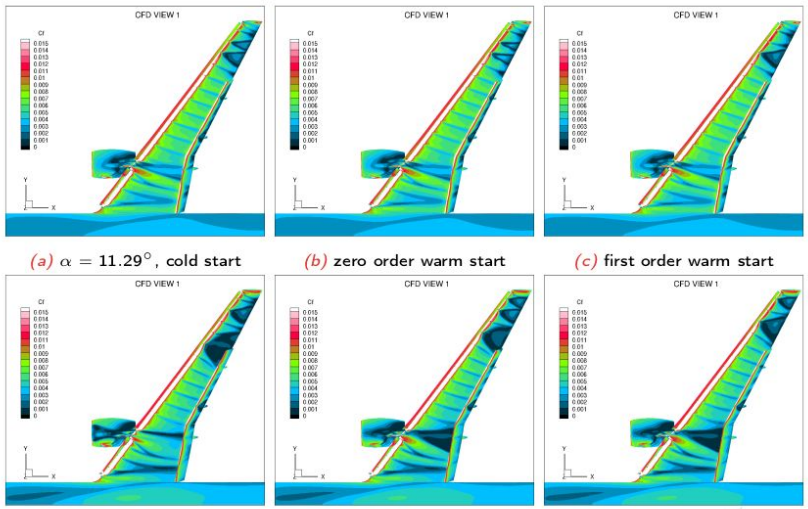
First-order warm start: initialize from the two solutions at the previous angles of attack (o: old, oo: older):

$$\vec{W} = \vec{W}_o + \frac{\vec{W}_o - \vec{W}_{oo}}{\alpha_o - \alpha_{oo}} (\alpha - \alpha_o)$$



A zero-order warm start simply takes the solution at the previous angle of attack as the initial condition.

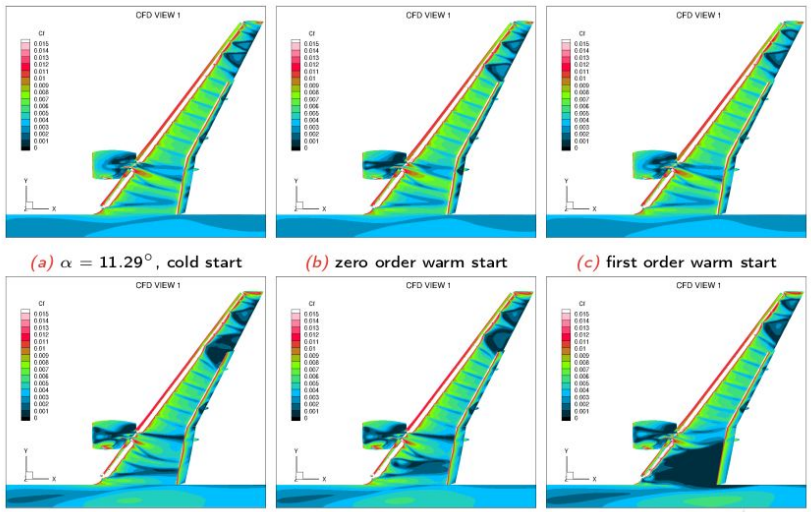
Warm restart



(a) $\alpha = 11.29°$, cold start (b) zero order warm start (c) first order warm start

(d) $\alpha = 17.05°$, cold start (e) zero order warm start (f) first order warm start

Surface friction coefficients Increasing α.



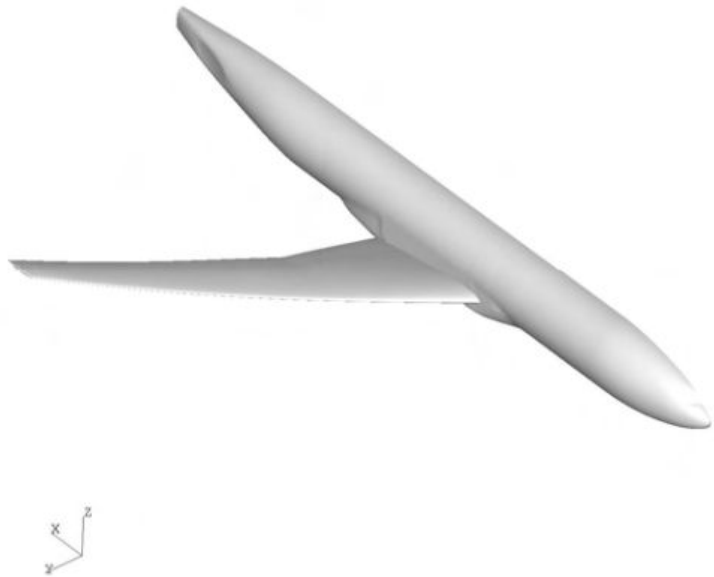
(a) $\alpha = 11.29°$, cold start (b) zero order warm start (c) first order warm start

(d) $\alpha = 17.05°$, cold start (e) zero order warm start (f) first order warm start

Surface friction coefficients Decreasing α.

Verification and validation: DPW7

The Drag Prediction Workshops targeted cruise drag prediction, but the 7th edition targeted higher angles of attack with shock-induced separation.





Ref: Tinoco, E. (2023). Summary data from the seventh AIAA CFD drag prediction workshop. In *AIAA AVIATION 2023 Forum* (p. 3492).

Verification and validation: DPW7 - fixed C_L

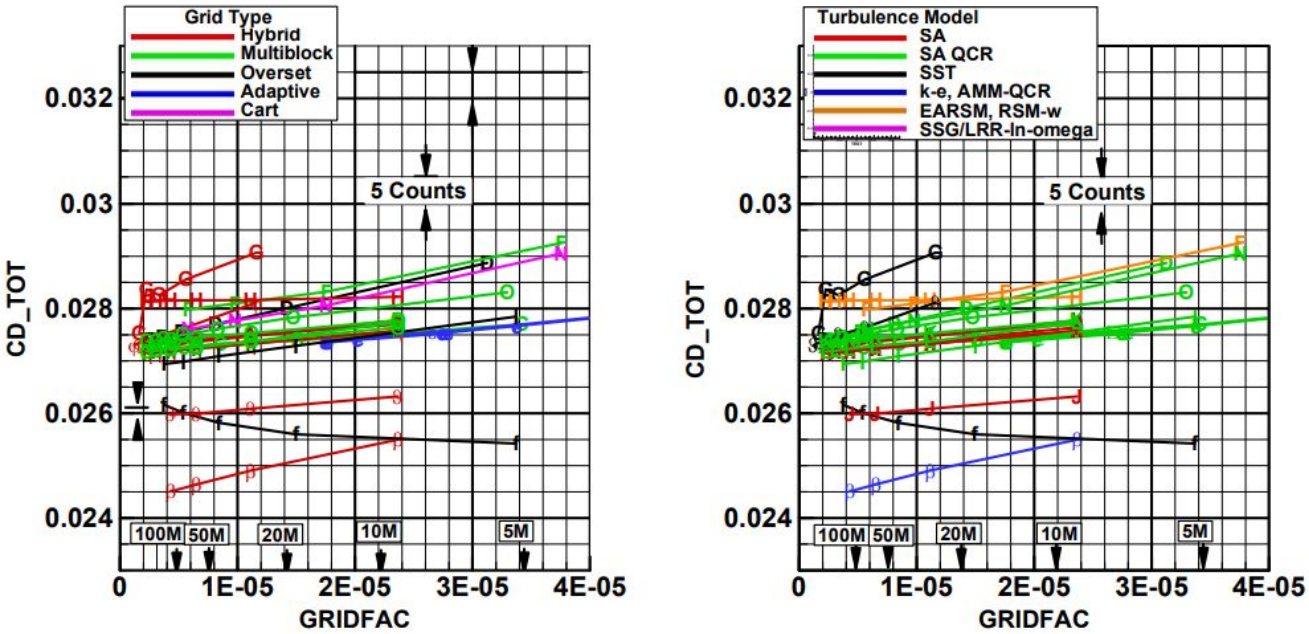


Fig. 10. Case 1a: Total drag by grid type and turbulence model, M=0.85, CL=0.58, Re=20 million.

Ref: Tinoco, E. (2023). Summary data from the seventh AIAA CFD drag prediction workshop. In *AIAA AVIATION 2023 Forum* (p. 3492).

Verification and validation: DPW7

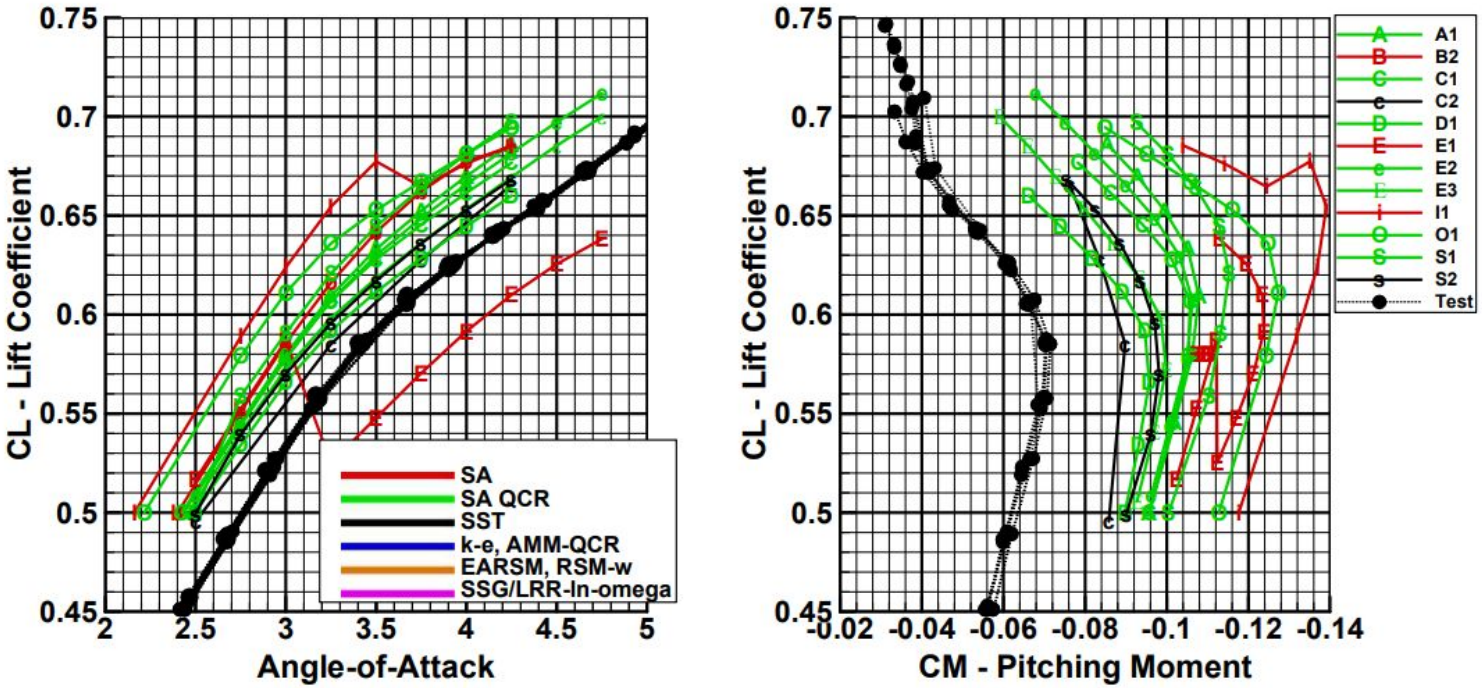


Fig. 28. Lift and pitching moment for all solutions, M=0.85, Re=5 million

Ref: Tinoco, E. (2023). Summary data from the seventh AIAA CFD drag prediction workshop. In *AIAA AVIATION 2023 Forum* (p. 3492).

Verification: DPW7

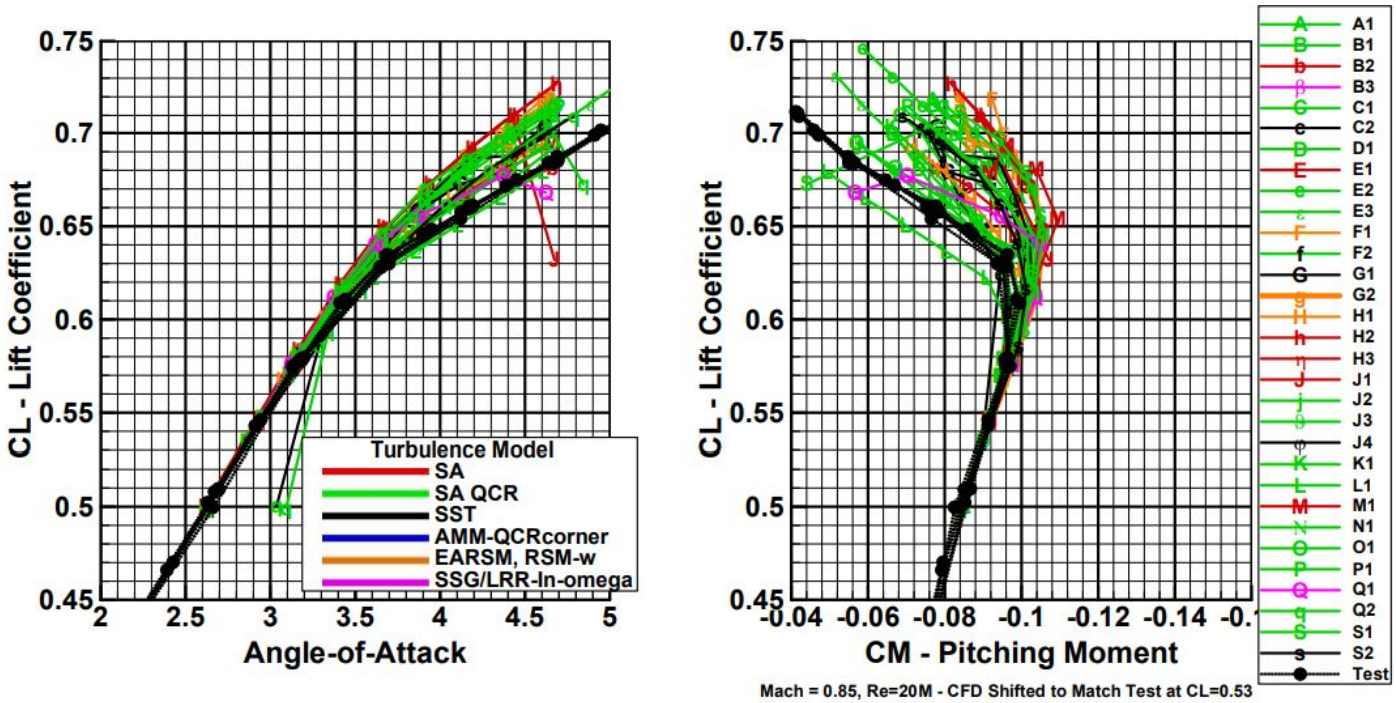


Fig. 38. Lift and Pitching Moment Shifted to Match Experiment at CL = 0.53, Chord Re=20 million.

Ref: Tinoco, E. (2023). Summary data from the seventh AIAA CFD drag prediction workshop. In *AIAA AVIATION 2023 Forum* (p. 3492).

Verification: DPW7

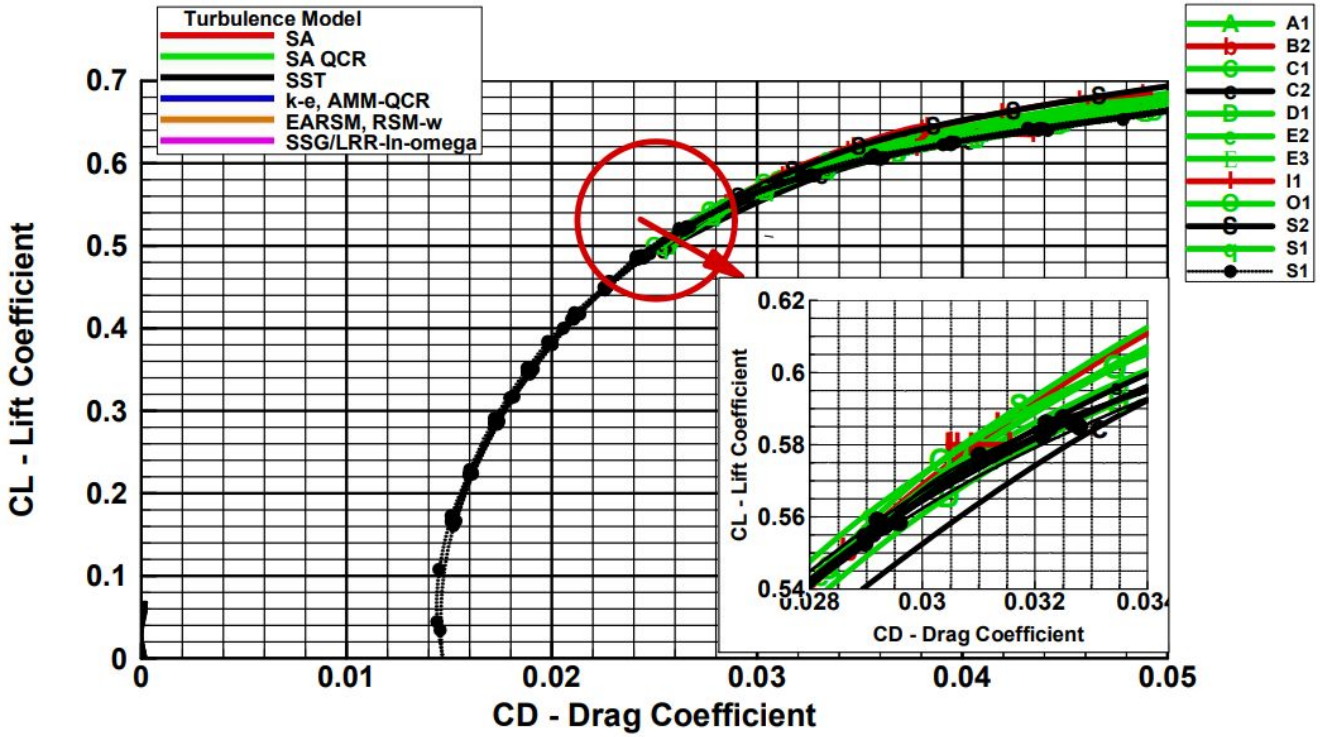


Fig. 29. Idealized Drag, M=0.85, Re=5 million.

Ref: Tinoco, E. (2023). Summary data from the seventh AIAA CFD drag prediction workshop. In *AIAA AVIATION 2023 Forum* (p. 3492).

Time-accurate simulation

Unsteady simulations

We have discussed acceleration techniques. Most of them rely on sacrificing the temporal accuracy (local time stepping, residual smoothing, multigrid, etc.). Even multistage schemes are designed for convergence rate and not for accuracy. How can we obtain a time-accurate simulation?

1. Use a time-accurate scheme with global time stepping (and be limited by the CFL limit);
2. Use dual time stepping;
3. Use a frequency-domain method.

Dual time stepping

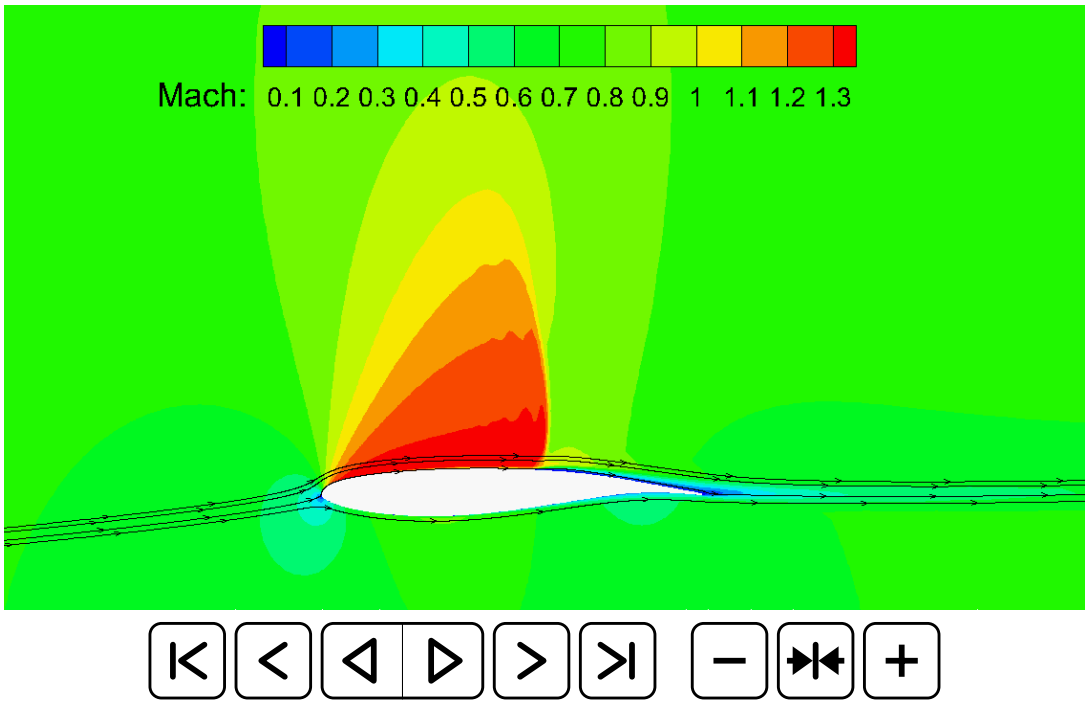
$$\Omega \frac{\partial U}{\partial t^\star} + \underbrace{\Omega \frac{\partial U}{\partial t}}_{\text{source term}} + \underbrace{R}_{\text{sum of the fluxes}} = 0$$

$$\Omega \frac{\partial U}{\partial t} \approx \frac{\Omega}{2\Delta t}\left(3U^{n+1} - 4U^n + U^{n-1}\right)$$

$\Delta t^\star$ is the local (pseudo) time step and Δt is the global (physical) time step.

We discretize the real time derivative with a second-order backward finite difference. So each time step becomes a "steady-state" problem. Every acceleration technique can then be used to compute the conserved quantities U at the next time step.

2D transonic buffet - RANS simulation



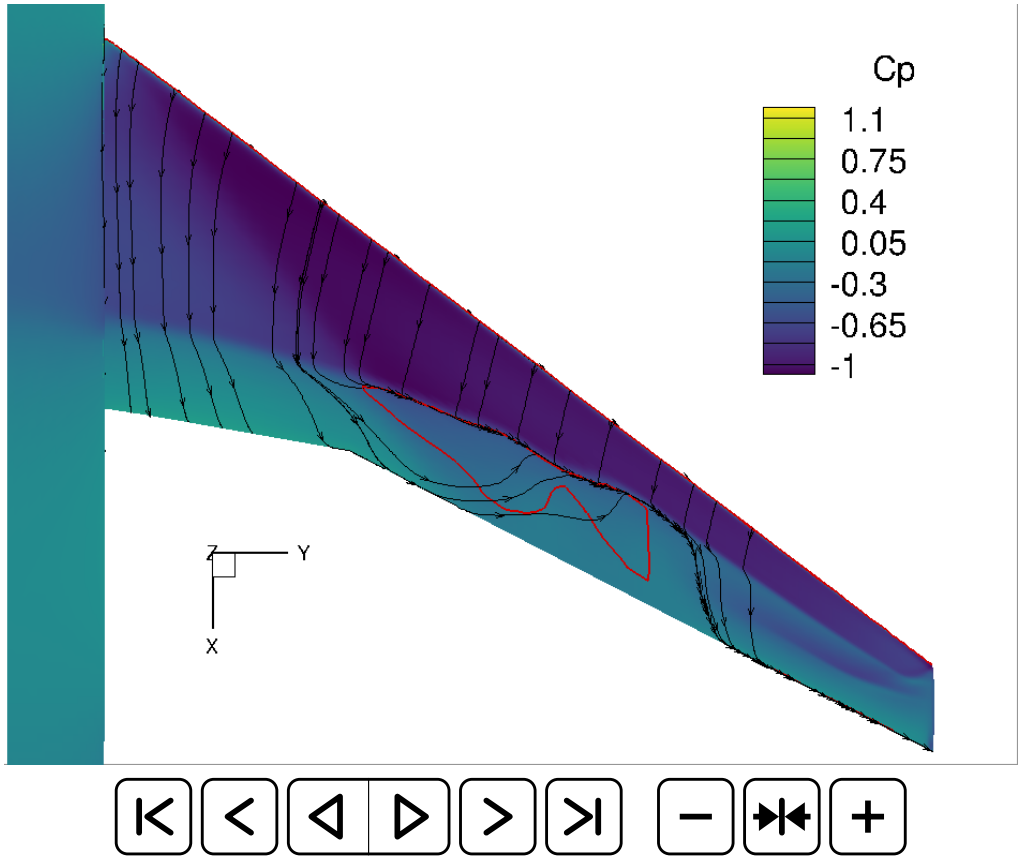
Transonic buffet:

- OAT15A;
- $M = 0.73$;
- $\mathrm{Re} = 3$ million;
- AoA $= 3.5$ deg.

Buffet is a self-sustained instability.

Ref: Plante, F. (2017). *Simulation de phénomènes instationnaires périodiques avec une méthode RANS dans le domaine fréquentiel* [Mémoire de maîtrise, École Polytechnique de Montréal]. PolyPublie. https://publications.polymtl.ca/2528/

3D transonic buffet - RANS simulation



3D transonic buffet:

- NASA CRM;
- $M = 0.85$;
- AoA $= 3.75$ deg;
- $\mathrm{Re} = 20$ million.

Ref: Plante, F. (2023). DPW7 results from CHAMPS, including global stability analyses. In *AIAA AVIATION 2023 Forum* (p. 3399).

Time spectral method (TSM)

Only for periodic cases

$$E^{-1} = \begin{bmatrix} 1 & \cos(\omega t_1) & \cdots & \cos(N\omega t_1) & \sin(\omega t_1) & \cdots & \sin(N\omega t_1) \\ 1 & \cos(\omega t_2) & \cdots & \cos(N\omega t_2) & \sin(\omega t_2) & \cdots & \sin(N\omega t_2) \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 1 & \cos(\omega t_{2N+1}) & \cdots & \cos(N\omega t_{2N+1}) & \sin(\omega t_{2N+1}) & \cdots & \sin(N\omega t_{2N+1}) \end{bmatrix}$$

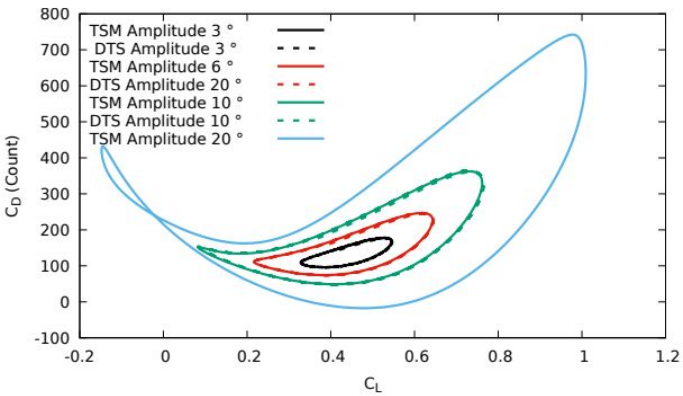
$$U = E^{-1}\hat{U}, \qquad \hat{U} = EU$$

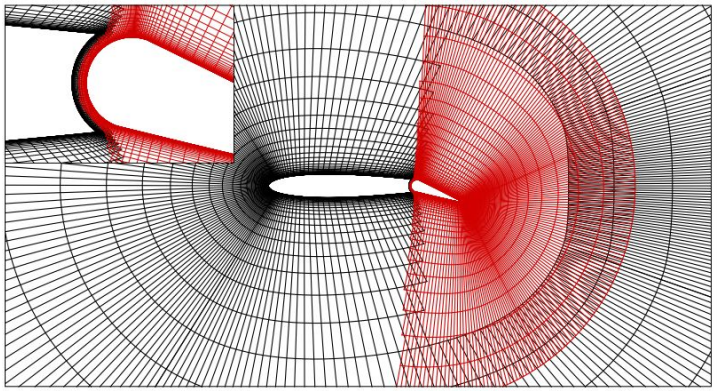
$$\frac{\partial U}{\partial t} = \omega D U = \frac{\partial E^{-1}}{\partial t} E U$$

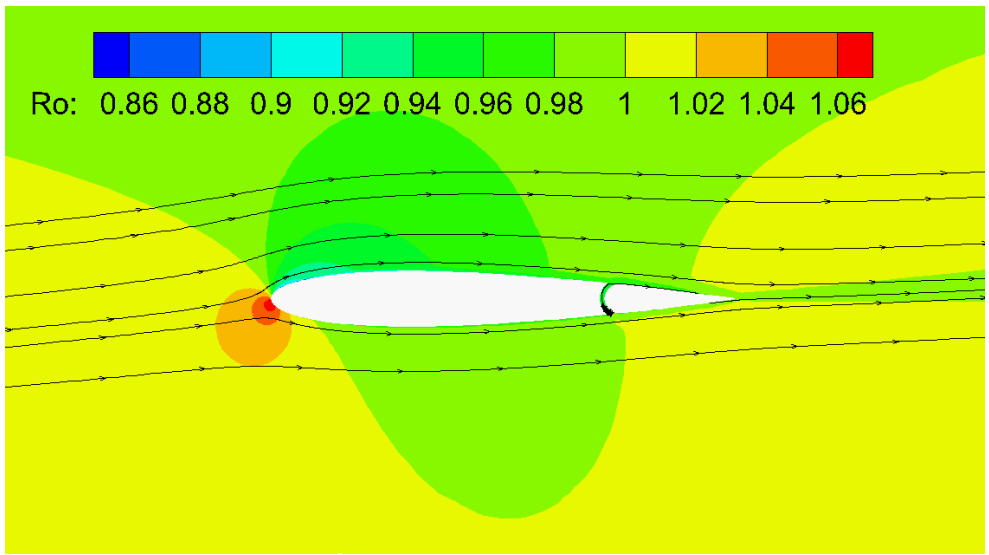
We use Fourier series to evaluate the global time derivative.

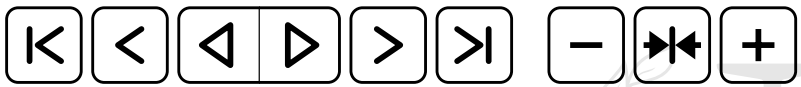
$$\Omega \frac{\partial U}{\partial t^\star} + \Omega\omega \underbrace{D(U)}_{\substack{\text{frequency-domain} \\ \text{time-derivative operator}}} + R = 0$$

Oscillating aileron - overset grid









Ref: Plante, F. (2017). *Simulation de phénomènes instationnaires périodiques avec une méthode RANS dans le domaine fréquentiel* [Mémoire de maîtrise, École Polytechnique de Montréal]. PolyPublie. https://publications.polymtl.ca/2528/